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Assignment-1 (Mid-term), Autumn-2025

Course: **CSE-3521 (Computer Architecture)**

Marks: **10**

[Answer all **three** questions; Figures in the right hand margin indicate Marks]

Q1.	Consider the following C statements with five integer variables a, b, c, d, e: $a = b + c;$ $d = a - e;$ Translate the equations in MIPS assembly instructions. Calculate the Machine code. Assume, a, b, c, d, e will be $\\$t0$, $\\$t1$, $\\$s0$, $\\$s1$, $\\$s2$. Explain: 1. Suppose the variables a and e are stored in the same memory location by mistake. Explain what happens when these instructions execute. 2. Write the expected values of a and d after execution in this scenario.	5	CO1 CO2
Q2.	The <i>first version of multiplication hardware</i> multiplies two integers x and y by performing repeated addition: adding x to an accumulator y times. A student decides to reuse the accumulator memory for both x and z to save hardware registers. Draw the flowchart. Explain: 1. Without computing exact numbers, explain why the result of $z = x * y$ may be incorrect in this scenario. 2. Describe a step-by-step sequence of additions showing how the accumulator changes if $x = 4$ and $y = 3$. 3. Suggest a creative fix that allows the hardware to still use minimal registers but always produces the correct result.	5	CO2 CO3